

Ayman El Didi

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Backend-leaning full-stack engineer building reliability-critical retail/rental systems with Go + React and deep Light-speed R-Series integration. Focused on scalable system design, near-real-time data sync, concurrency, failure handling, and measurable performance improvements.

WORK EXPERIENCE

Software Developer, Med Supplies (Vine Media inc.)

Edmonton, Alberta Nov 2024 — Present

- Technical owner for a mission-critical rental + inventory platform with **Lightspeed R-Series** integration supporting real-time, location-based availability and operational workflows (Go, ASP.NET Core, React, MySQL, Redis).
- Designed **near-real-time ingestion + reconciliation** from R-Series (streaming sync, backfills, and consistency checks), meeting a **2-minute worst-case** propagation SLA for upstream changes.
- Built multi-tenant worker system with strict timeouts, retries/backoff, and an ops dashboard (health, last-error, restart controls) to harden reliability under third-party failures (rate limits, outages, partial degradation).
- Implemented idempotent workflows and correctness guards (pending states, post-sale confirmation, anomaly detection for void/refund drift) to keep rental state consistent with POS-side actions.
- Authored typed R-Series client + distributed Redis token-bucket limiter aligned to upstream headers; stabilized concurrency and sustained **7+ months incident-free** operation while collaborating with Lightspeed product/engineering on API feedback.

Lead Software Developer (Concurrent Contract Venture), ATG Sports

Edmonton, Alberta (Remote) Jan 2025 — Nov 2025

- Led a 5-person team delivering a scouting/ops platform; owned architecture and major components across back-end + UI (ASP.NET Core, React/React Native, PostgreSQL, Redis, Render.com).
- Drove performance initiative: **21% overall improvement, 5x speedups** on critical endpoints by reshaping query patterns, adding more finely-partitioned caching, and materializing search-optimized views (OpenSearch + Redis) with controlled staleness.
- Built ingestion pipelines and operational tooling; partnered directly with founder/PM on roadmap tradeoffs and delivery; received formal letter of recommendation.

Software Developer - Go (Contract work done during school), Berry Homes

Edmonton, Alberta (Remote) May 2023 — Sept 2023

- Developed a Go web service to ingest company data and provide visualizations and metrics in a dashboard which remains in use internally

TECHNICAL SKILLS

- **Languages:** Go, Typescript/Javascript, C#, Rust, C, C++, Python, Bash/POSIX Shell
- **Backend & APIs:** Go (Standard Library), ASP.NET Core, Axum, Flask, RESTful APIs, distributed systems
- **Web & Mobile:** React, React Native, Expo
- **Data:** PostgreSQL, MySQL, Redis, OpenSearch
- **Infrastructure & Tooling:** Linux, Docker, Docker Compose, Nix, VPS Administration, Playwright, Git, GitHub, CI/CD, Nginx, Caddy, Lightspeed R-Series API

OPEN SOURCE, SYSTEMS & PERFORMANCE RESEARCH

Unicode ID Trie RLE Compression & Benchmarking (Go & Rust) .. <https://github.com/aeldidi/unicode-id-trie-rle>

- Implemented compressed Unicode ID trie with **35% less static storage** than leading library with competitive performance; ported to C and Go
- Performed cache-aware microbenchmarks across multiple datasets
- Authored a detailed technical writeup: <https://ayman.eldidi.org/articles/unicode-identifiers>

AlbumArt.digital — High-Resolution Album Art Search Engine (Go) <https://albumart.digital>

- Source code: <https://github.com/aeldidi/albumart.digital>
- Dependency-free Go web service integrating directly with Apple Music API for high-resolution album art

EDUCATION

University of Alberta BSc Double Major in Computing Science and Mathematics, Sept 2019 — Apr 2024